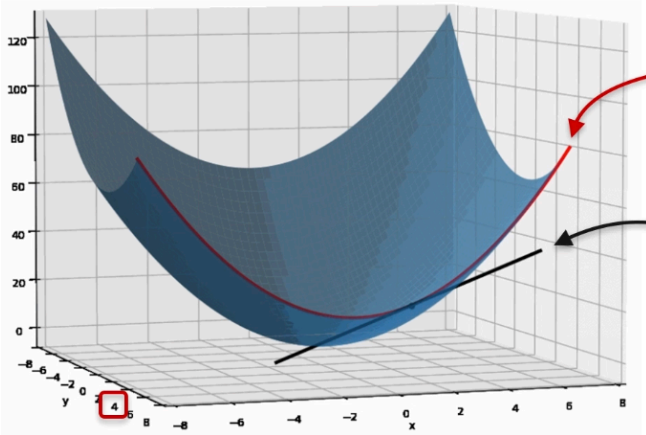
Partial Deravatives

$$f(x, y) = x^2 + y^2$$

Treat y as a constant



Constant

$$f(x, y) = x^2 + \boxed{y^2}$$

$$\frac{\partial f}{\partial x} = 2x + 0$$

$$x^2 + y^2$$

Treat y as a constant → $2x$

Treat x as a constant → $2y$

$$f(x, y)$$

$$f_x = \frac{\partial f}{\partial x}$$

Partial derivative of f with respect to x

$$f_y = \frac{\partial f}{\partial y}$$

Partial derivative of f with respect to y

$\partial f = \text{di of } f$